

# AWS S3 – Enable Versioning and Data Recovery

## Objective:

To protect data using S3 Versioning and understand how to recover deleted or modified objects.

## (a) Create S3 Bucket:

1. Login to AWS Management Console  
Open <https://aws.amazon.com/> and sign in
2. Open **S3 Service**  
Search and select S3
3. Click **Create bucket**
  - Bucket Name: my-versioning-bucket-12345 (must be unique)
  - Region: choose nearest region
4. Click **Create bucket**

## (b) Enable Versioning:

1. Open created bucket
2. Go to **Properties tab**
3. Find **Bucket Versioning**
4. Click **Edit**
5. Select **Enable**
6. Click **Save changes**

## (c) Upload File (Version 1):

1. Go to **Objects** tab
2. Click **Upload**
3. Add file  
Example: `file.txt` (content: Hello World)
4. Click **Upload**

## (d) Modify and Upload Same File (Version 2):

1. Edit same file locally  
Example: change content to `Hello AWS Version 2`
2. Upload same file again
3. AWS automatically creates a **new version**

## (e) View Object Versions:

1. Open bucket
2. Enable **Show versions** option
3. You will see:
  - Version 1
  - Version 2

## (f) Download Older Version:

1. Select file
2. Choose **Version 1**
3. Click **Download**
4. Old content is restored locally

## (g) Delete Object:

1. Select file
2. Click **Delete**
3. AWS does NOT permanently delete (creates delete marker)

## (h) Restore Deleted Object:

1. Enable **Show versions**
2. Select delete marker
3. Click **Delete delete marker**
4. Object is restored

## (i) Permanently Delete a Version:

1. Enable **Show versions**
2. Select specific version (Version 1 or 2)
3. Click **Delete**
4. This removes version permanently

## Result:

- Versioning enabled successfully
- Multiple versions of file created
- Older versions restored successfully
- Deleted objects recovered using versioning
- Permanent deletion done at version level
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